

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Factorising

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

FACTORISING · CH2 · L4

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The Map

This lesson collapses Factorising into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Take out the common factor

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Factorise a quadratic

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use difference of squares

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Check by expanding

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Factorise expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 (b)(i)

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 23 website questions, 23 checked solutions, 3 local PDFs read.

01 Take out the common factor

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Factorising question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q9

(a)

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

(b)(i)

$$x^2 + 5x - 36 = (x + 9)(x - 4)$$

Finish: (a) $16x^{12}y^{20}$, (b)(i) $(x + 9)(x - 4)$, (b)(ii) $x = -9$ or $x = 4$

FORMULA BANK

Look for common factor first, then brackets, then difference of squares or quadratic factorising.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Factorise a quadratic

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Factorising question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

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SAME IDEA Use the same first line from Strategy 02.

QUESTION

9 (a) Simplify $(2x - 3y + 5)^4$ (2) (b) (i) Factorise $x^2 + 5x - 36$ (2) (ii) Hence, solve $x^2 + 5x - 36 = 0$ (1)

YOUR SOLUTION

03 Use difference of squares

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Factorising question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$n^2 - 7n + 12 = (n - 3)(n - 4)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1HR Q9

(a)

This is a reasoning step: write one clear sentence, then continue the calculation.

(b)

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) x^9 , (b) $64y^6$, (c) $(n - 3)(n - 4)$

FORMULA BANK

Look for common factor first, then brackets, then difference of squares or quadratic factorising.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

9 (a) Simplify x^4 times x^5 (1) (b) Simplify $(4y^2)^3$ (2) (c) Factorise $n^2 - 7n + 12$ (2)

04 Check by expanding

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Factorising question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$9x^2 - 4y^2 = (3x)^2 - (2y)^2 = (3x - 2y)(3x + 2y)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1H Q11

(a)

$$9x^2 - 4y^2 = (3x)^2 - (2y)^2 = (3x - 2y)(3x + 2y)$$

(b)

$$\frac{7}{8} - \frac{x+3}{4x} = \frac{7x}{8x} - \frac{2(x+3)}{8x}$$

Finish: (a) $(3x - 2y)(3x + 2y)$, (b) $\frac{5x-6}{8x}$

FORMULA BANK

Look for common factor first, then brackets, then difference of squares or quadratic factorising.

BOTTOM LINE

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SAME IDEA Use the same first line from Strategy 04.

QUESTION

11 (a) Factorise $9x^2 - 4y^2$ (2) (b) Express $\frac{7}{8} - \frac{3}{4} + \frac{1}{x} + \frac{1}{x}$ as a single fraction in its simplest form. (3)

YOUR SOLUTION

05 Factorise expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Factorising question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

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WORKED MODEL

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Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
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"work out", "show", "hence", a familiar exam phrase	Factorise a quadratic
"work out", "show", "hence", a familiar exam phrase	Use difference of squares
"work out", "show", "hence", a familiar exam phrase	Check by expanding
"work out", "show", "hence", a familiar exam phrase	Factorise expression
"work out", "show", "hence", a familiar exam phrase	(b)(i)

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.